

Mini Project 6: HAZOP Bebop

Che 396 - Senior Design I

December 7th, 2019

Instructor: Dr. Betul Bilgin

James Fell

Jasmine Villegas

Mariana St. Peter

William Rawson

Table of Contents

1.0	Project Charter	3
2.0	Design Plan	4
3.0	R-100 and T-100 Process Unit	5
3.1	Environmental Hazards and Impacts from the Direct Chlorination Reactor (R-100) and the HCl Column (T-100)	5
3.2	Inherently Safer Design of the reactor (R-100) and column (T-100)	7
3.3	Process Hazards	8
3.4	OSHA PSM Highly Hazardous Materials	11
3.5	HAZOP Analysis of the Vinyl Chloride Process	11
4.0	References	22

1.0 Project Charter

Title	Goal	In Scope	Out of Scope
Mini Project 6: Safety and Environmental Hazards of Vinyl-Chloride Process	Th goal of this project is to investigate the safety and environmental hazards of vinyl-chloride production	Environmental issues Mitigation of environmental issues Process hazards Analysis of process safety with HAZOP study	OSHA PSM list for hazards materials Safeguard for the process Analysis of safety with HAZOP for compounds involved in the production of vinyl-chloride
Deliverable		Team Members	
• List of environmental issues • Mitigation of issues listed • Ethical and legal relation of issues • Analysis of R-100 and T-100 process units		James Fell Jasmine Villegas Mariana St. Peter William Rawson	

2.0 Design Plan

Project Planner

Select a period to highlight at right. A legend describing the charting

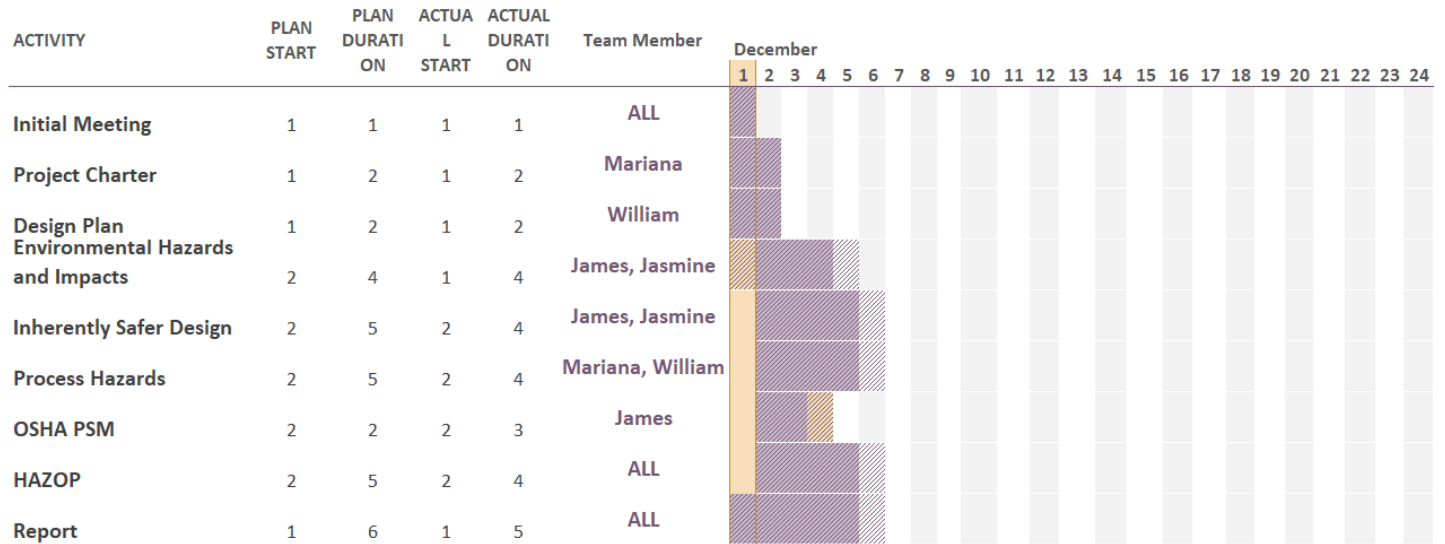
Period Highlight: 1

Plan Duration

Actual Start

% Complete

Actual (beyond pl.



3.0 R-100 and T-100 Process Unit

3.1 Environmental Hazards and Impacts from the Direct Chlorination Reactor (R-100) and the HCl Column (T-100)

In the direct chlorination reactor (R-100), the materials that pass into and are created in the reactor are ethylene, chlorine gas, and 1,2-dichloroethane. The direct chlorination reactor reacts ethylene and chlorine in an exothermic reaction to produce 1,2-dichloroethane. The reactor is maintained at a temperature of 90°C by using a cooling jacket with a cooling water. Ethylene and chlorine are in the gas phase going through the reactor and 1,2-dichloroethane is a liquid leaving the reactor. This presents a major concern because these chemicals are corrosive. The possibility of chemical spills and leaks into the cooling water, manufacturing facility's air, and outside air is very high. A brief description of the environmental concerns for ethylene, chlorine, and 1,2-dichloroethane should a spill or leak occur are described below:

- Ethylene itself is not that dangerous to the environment nor does it pose significant health risks [3]. Elevated levels of Ethylene are commonly found in the environment and have not shown to damage or harm wildlife or plants [5]. Ethylene also does not pose a significant water hazard as its solubility in water at 90°C is approximately 0 and is relatively nontoxic [6]. The major concern for ethylene is its flammability. The flammability limit for ethylene is low (~3%), which can create a significant explosion hazard even if a small amount of ethylene escapes into the air and finds an ignition source.
- 1,2-Dichloroethane itself is not that dangerous to the environment nor does it pose significant toxicity risks, but it is suspected of being a carcinogen with chronic exposure [3]. If large spills of 1,2-dichloroethane occur, it can harm wildlife and contaminate water bodies. It does not bind to soils or sediments which means it can accumulate and persist for many years in bodies of water [5]. Since 1,2-dichloroethane is a volatile organic compound, if a spill occurs in the facility, the liquid will vaporize readily, exposing the workers to 1,2-dichloroethane above the permissible exposure limit (PEL) and increases the risk of a fire/explosion hazard.
- Chlorine itself is dangerous to the environment and poses significant health risks as it is highly corrosive and carcinogenic [3]. Chlorine has shown to cause environmental harm, even at low levels. Chlorine is particularly harmful to organisms living in the soil or water [5]. Due to the high corrosiveness of chlorine and the gaseous nature of it, the risk of leaks into the air and cooling water are high if pipes/pipe welds are corroded. Even a small leak of chlorine gas into the facility can easily expose everyone to more than the PEL for chlorine. Also, chlorine does pose a significant water hazard as chlorine gas is soluble in water at 90°C [6]. If this contaminated water goes into environmental or municipal water, ecosystems and human health can be affected by the contamination.

In the HCL distillation column (T-100), the materials that pass into the distillation column are 1,2-dichloroethane, vinyl chloride, and hydrogen chloride. The purpose of this distillation column is to separate the hydrogen chloride from the vinyl chloride and 1,2-dichloroethane. The vinyl chloride will be separated from the 1,2-dichloroethane in another distillation column to produce pure vinyl chloride. The column runs at a condenser temperature of -26.4°C and a boiler temperature of 94.6°C. The major concern for the column is that the chemicals are corrosive and some are toxic. The risk of chemical spills and leaks into the medium pressure steam, manufacturing facility's air, and outside air is high due to the high volatility of the chemicals. A brief description of the environmental concerns for 1,2-dichloroethane, vinyl chloride, and hydrogen chloride should a spill or leak occur are described below:

- 1,2-dichloroethane was discussed above. The same hazards that were addressed for the reactor also apply to the distillation column.
- Release of vinyl-chloride can increase the risk of fire or explosion hazards. In case of spill, isolate the area and deny entry of all unprotected personal. All sources of heat and ignition must be removed from the area since this compound presents a low flammability level of 3.6% for the lower level and 33.0% for the upper level as it shows in **Table 1** below. This compound is believed to be non-toxic to aquatic life on an acute basis, which correspond to LC₅₀ greater than 100mg/L. In case of a released to the atmosphere, vinyl-chloride will remain in the gas phase, volatilization will occur if released to soil and evaporation will occur if released to water.
- Hydrogen chloride can cause corrosivity of metal and limestone structures. If high amounts of hydrogen chloride are released into a body of water, organisms living in said body of water can become ill or deceased. Because hydrogen chloride is highly soluble, release into the atmosphere is washed out by rain and moisture in the air. However, "critical loads" can be reached in soil or lakes if acid rain caused by the absorption of released hydrogen chloride reaches the allotted threshold. Therefore, hydrogen chloride pollution can be considered a global and local environmental problem [5].

3.2 Inherently Safer Design of the reactor (R-100) and column (T-100)

Caution must be taken when working with hazardous chemical compounds inside the plant or when operating a process that poses risks to the surroundings. Simple routine training, preventative maintenance, as well as well-organized operating manuals and emergency procedures should be applied in order to evaluate the process and ensure the integrity of all equipment. It is also important the awareness of all the employees about the chemicals been used and their properties, this can avoid hazardous operations.

Although many of the environmental issues described above have already been addressed by process controls added to the system in order to produce vinyl-chloride safely, a couple extra precautions can be taken into consideration.

R-100

A safer design for the reactor first starts with a highly chemical resistant material of construction. From previous design work of the reactor, the safest material to use with the chemicals being processed was 316 stainless steel. Along with the materials of construction, the following schedules and devices can be implemented to reduce hazards to the facility, workers, and the environment:

- Regularly scheduled maintenance on the pipes, shells, and tubes that are in contact with the chemicals.
- Regularly scheduled checks for leaks around pipe fitting and welds. Repairs need to be made immediately if any anomalies are noticed.
- Proper ventilation in the facility.
- Sample the cooling water periodically for chlorine, ethylene, and 1,2-dichloroethane contamination.
- The pressure relief system has a scrubber to remove any chlorine or 1,2-dichloroethane before it goes into the environment.
- All ignition sources need to be isolated away from the equipment with adequate ventilation around the ignition sources. All electrical boxes need to be explosion proof.

T-100

A safer design starts with choosing the right material for the vessel, as well as the right conditions of operation. As previously analyzed in Project 3, the material chosen for the vessel was Incoloy, which would be able to prevent corrosion from the hydrogen chloride. In addition to the material and operating conditions, the following can be applied:

- Regular scheduled maintenance on the pipes
- Regular scheduled checks for leaks around pipe fitting and welds
- Sample the medium pressure steam periodically for 1,2-dichloroethane, vinyl chloride, and hydrogen chloride contamination
- Isolate any form of ignition in case a leakage happens in the column

3.3 Process Hazards

Dispose of all the compounds and containers must be done in accordance with all local, regional, national and international regulations. The following are a list of process hazards for the production of vinyl-chloride.

Loss of Containment

This hazard is related with the protection of the community around the plant. Toxic chemicals must be contained by the plant itself to secure the safety of the public.

In case of accidental release of toxic or explosive compounds, the consequences can be disastrous and need to be safeguarded against to protect the workers, the facility and the community. With that in mind, continuous air monitoring around the chemical plant is

crucial, where chemical levels should be kept below the permissible exposure limit (PEL) and when possible outside of the flammable limits (LEL and UEL). In case of an accidental release that would result in damages to wildlife, water lines and the community, authorities should be informed right away, and emergency procedures should take place.

Emergency procedures include the elimination of all ignition source if safe to do so, do not extinguish the leak unless it can be repaired safely. In case of eye contact, immediately flush eyes with clean water for at least 15 minutes, remove any contact lenses and seek medical attention. In case of inhalation, move person to fresh air and allow the person to rest in a comfortable position. If victim is not breathing, provide artificial respiration, do not provide mouth-to-mouth resuscitation, since this technique may put you at risk of contamination. Get medical attention. In case of skin contact, flush contaminated area with clean water for at least 15 minutes, remove contaminated clothes and shoes after soaking them with water to avoid static discharge been an ignite source. Get medical attention. In case of ingestion, remove victim to open fresh air area, remove dentures if any and wash victim's mouth with water. Get medical attention. In case of frostbite, try to warm up the tissues that have been damaged, and seek medical attention.

In case of a fire, promptly isolate the surrounding areas. Shut off all ignition sources. Avoid breathing gas.

Pressure

When the pressure applied exceeds the maximum safety pressure range allowed by the design specifications, the system is said to be overpressure. In case of an overpressure, a failure of the equipment may occur, this usually occur at the weakest part of a join or weld.

During a process control setup, safety pressure release valves should be installed in closed vessels, where is possible to release the pressure in case of an over pressurized situation occurs. Explosions from buildup pressure is a serious hazard that can be encounter in a chemical facility.

Temperature (and autoignition)

Temperatures above the maximum safety temperature range allowed by the design specifications can result in failure of the equipment. High temperatures can also consequence in high pressure inside a closed vessel, which may lead to catastrophic disasters. High temperatures can occur by failure of a cooling system, as well as the loss of control of a reaction or heat source.

In order to avoid this type of issue, temperature elements that signal a control box must be installed in the system. Such equipment would be able to shut down reactor feeds, heating sources or even increased the flow in cooling system. It is also important to note that the temperature must be kept below 440°C to avoid any autoignition of the flammable compounds, such as ethylene, 1,2-dichloride and vinyl-chloride. Due to safety and cost efficiency, steam heaters are the preferable source of heat in a chemical plant,

which limits the temperature rise of the system when compared to fired heater. On the other hand, low temperatures can also be hazardous to the process. Since pipes can freeze and crack, jeopardizing the safety of the plant with possible chemical leakage.

Material Incompatibility

This is one of the major sources of incidents in a chemical plant. There are two main incompatibility of material that must be taken very seriously. The first is the incompatibility of the chemicals with the material of construction, which means that the chemicals been used in a determinate equipment can lead to loss of matter. Such damage can be summarized as the corrosion of the vessels been used, as well as the instruments, softening of gaskets, seals and lining, which would result in leaks. The second one is the incompatibility of the chemicals been mixed, which could result in a violent reaction such as extreme exothermic reactions or temperature runaway. For the production of vinyl-chloride, the compatibility of the chemicals involved in the process was checked by using the CRW4 software, see the **Figure 1** below.

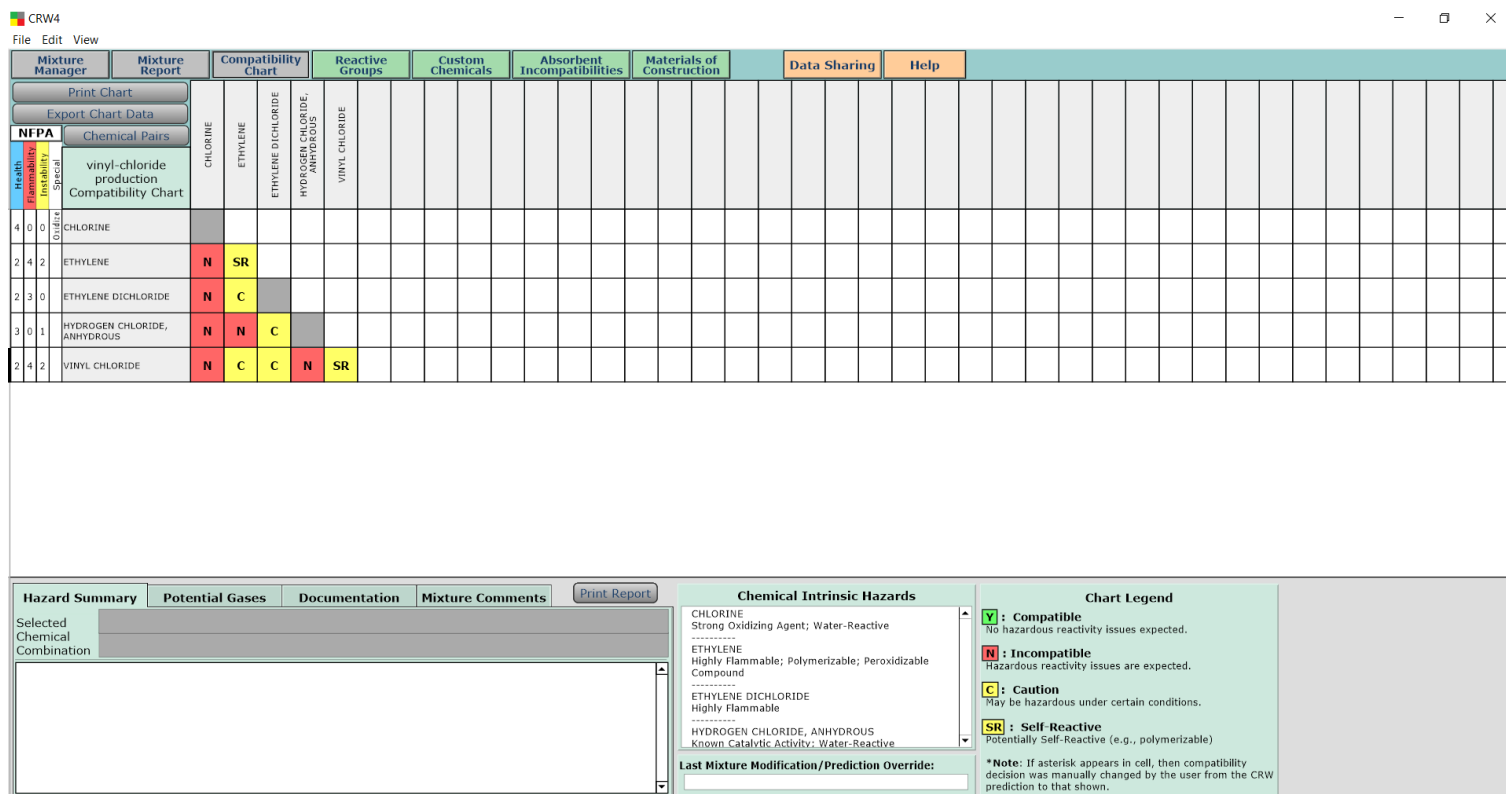


Figure 1: Compatibility chart for the chemicals involved in the production of vinyl-chloride

As **Figure 1** shows, all the chemicals are incompatible with chlorine, which means that the reaction must be done very carefully since hazards reactivity is expected. Ethylene and vinyl-chloride show as self-reactive compounds which potentially could result in an unexpected reaction.

Fire and Sources of Ignition

A fire can occur when three components are present: fuel, source of ignition and an oxidizing agent (usually oxygen). In case the temperature is above the autoignition temperature of the chemical, an ignition can happen spontaneously in the presence of air. For that reason, it is important to keep the temperature below this level, as described above. In case of a flammable material leak occurs, it is important to remove all sources of ignition such as:

- Electrical equipment: this is one of the main sources of ignition and can be avoided by explosion proof enclosures, pressurized and purged equipment and also by non-sparking equipment. Simple routine training and preventative maintenance can minimize the chances of an electrical ignition source in addition to following the correct codes before selecting the equipment for a designated area.
- Static electricity: it is an electric charge at the surface of a material or at its surrounding. It can be generated by powder, liquid or gas, which will produce sparks. To avoid this kind of energy, pipes and equipment must be properly grounded.
- Open flame: known as process flame, this kind of ignition comes from furnaces, incinerators, heaters and other. This kind of ignition must be kept away from flammable materials and processes.

Noises

This kind of hazard jeopardize safety and health. The noise can be a result of compressors, turbines, motors and solid handling. Exposure to excessive noise for long periods of time can result in permanent damage hearing. Hearing protection should be a requirement for areas where the noise is greater than 80dB, since permanent loss of hearing occurs for noises above 85dB.

3.4 OSHA PSM Highly Hazardous Materials

Table 1. Identification of OSHA highly hazardous materials and their respective permissible exposure limits (PEL) [2,3].

Chemical	Highly Hazardous	OSHA PEL (ppm)
Ethylene	No	200
Chlorine	Yes	1
1,2-Dichloro Ethane	No	200
Vinyl Chloride	No	1
Hydrogen Chloride	Yes	5

Table 2. Identification of toxic and flammable chemicals used in the process. For flammable chemicals, the flash point, auto ignition temperature, and lower/upper flammability limits are noted [4].

Chemical	Toxic	Flammable	Flash Point (°C)	Auto Ignition Temperature (°C)	Flammable Limits (LEL, UEL)
Ethylene	No	Yes	-136.6	490	2.7 - 36%
Chlorine	Yes	No	-	-	-
1,2-Dichloroethane	Yes	Yes	13	440	6.2 – 15.9%
Vinyl Chloride	No	Yes	-78	472	3.6 – 33.0%
Hydrogen Chloride	Yes	No	-	-	-

3.5 HAZOP Analysis of the Vinyl Chloride Process

HAZOP is a hazard and operation study with details of a process design. This type of study is required by governmental regulations and was adopted by OSHA on May 26, 1992 [8]. HAZOP is dedicated to identifying potential hazards and operation problems within a system, which would be likely to result in nonconforming product [9]. It is composed of reviews of the P&ID section, called nodes, by specified groups such as process engineers, instrument engineer, HS&E expert, mechanical engineer, operations experts and another expert as needed [10].

HAZOP methodology is based on the following steps:

- **Definition**
Where objectives and scopes of the process are defined, as well as the responsibilities and the team responsible for those tasks.
- **Preparation**
Where the data is collected, and a plan of study is put in place. An arranged schedule and estimate time of completion are also discussed in this section.
- **Examination**
In this section, the system is divided into parts, which each part has a defined design. Deviations of the project, consequence, causes, problems, protection, detection and indicating mechanisms are also identified during this stage. Possible mitigation plan is also discussed. This step is repeated for each element of each part.
- **Documentation and Follow-up**

In this section, examination of the process is recorded, and documentation of the design are signed off and the report of the study is produced. Action that are implemented will be followed up and a re-study will be done in the system if necessary.

It is crucial to understand the guide words used in the HAZOP, and **Table 3** describe its meaning.

Table 3: Basic HAZOP guide word [10]

Guide Word	Meaning [10]
No (not, none)	None of the design intent is achieved
More (more of, higher)	Quantitative increase in a parameter
Less (less of, lower)	Quantitative decrease in a parameter
As well as (more than)	An additional activity occurs
Part of	Only some of the design intention is achieved
Reverse	Logical opposite of the design intention occurs
Other than (other)	Complete substitution – another activity takes place

For the purpose of this report, two nodes were studied: the direct chlorination reactor (R-100) and HCl Column (T-100).

HAZOP Study		Direct Chlorination Reactor (R-100)				
Design Intent:		React a gaseous chlorine feed stream and a gaseous ethylene feed stream in an exothermic reaction forming 1,2-dichloroethane. The reactor is cooled by cooling water flowing through the outside jacket in the reactor.				
Parameter	Guide Word	Deviation	Possible Causes	Consequences	Safeguards	Recommendations
Flow	No/ Less off	Reactants flow is reduced or stopped flowing	- Inlet flow leaked out due to pipes corroding, weld/fitting/joints loose or corroded - Inlet fluid pump stopped working due to mechanical problems/ corrosion/ age or cavitation - Pipes became blocked due to biomass build up, mineral build up or corrosion - Failure of control valve	- No properly reaction occurs - Decrease of product yield - Decrease of reaction conversion - Release of hazardous material - Increase of pressure inside pipes in case of a blockage - Decrease of pressure inside reactor	- Flow meters - Inspect lines and pump periodically - Installation of filters	- Install a flow indicator with low alarm (FAL) - Periodically inspection of lines, pumps and equipment for deterioration and corrosion - Install filter in the inlet line to prevent any solid blockage, check filters periodically
	More	Reactants flow is increased	- Failure of control valve - Pressurized pipelines - Failure in pump control system	- Reactor is overflowed - Reaction doesn't occur as desired - Increased of pressure inside reactor - Undesired product purity - Waste of raw material	- Flow meters - Inspect lines and pump periodically	- Install a flow indicator with high alarm (FAH) - Periodically inspection of lines, pumps and equipment
	Reverse	Reverse flow in pipelines	- Failure of control valves - Failure of pressure controller - Failure of pump	- Outlet flow of the reactor is in reverse direction - Decrease in product yield - Decrease in reaction conversion	- Flow meters - Inspect lines and pump periodically	- Install a flow indicator - Periodically inspection of lines, pumps and equipment

				Decrease in product purity		
	More of	Increase of cooling jacket fluid flow	- Failure of temperature indicator - Failure of flow indicator - Failure of control valve	- Decrease of temperature inside the reactor - Decrease of pressure inside the reactor - Reaction may not proceed as expected - Decrease of product yield - Decrease of reaction conversion - Increase of side products reaction - Increase of pressure inside pipeline of cooling system	- Temperature indicator in the reactor - Flow meters - Inspect lines and pump periodically	- Install a temperature indicator in the reactor - Install a flow indicator with high alarm (FAH) in the cooling system - Periodically inspection of lines, pumps and equipment for deterioration and corrosion
	No/ Less of	Cooling jacket fluid reduces or stops flowing	- Failure of temperate indicator - Cooling fluid leaked out due to pipes corroding, weld/fitting/joints loose or corroded - Cooling fluid pump stopped working due to mechanical problems/ corrosion/ age or cavitation - Pipes became blocked due to biomass build up,	- Increase of temperature inside the reactor - Increase of pressure inside the reactor - Run-away reaction - Decrease of product yield - Decrease of reaction conversion - Increase of side products reaction - Release of hazardous material - Increase of pressure inside pipes in case of a blockage	- Temperature indicator in the reactor - Flow meters - Inspect lines and pump periodically - Installation of filters - Emergency shut off of process - Safety pressure valve release - Evacuation alarm	- Install a temperature indicator in the reactor - Install a flow indicator with low alarm (FAL) in the cooling system - Periodically inspection of lines, pumps and equipment for deterioration and corrosion - Install filter in the inlet line to prevent any solid

			mineral build up or corrosion • Failure of control valve	• Increase rate of corrosion • Autoignition of ethylene and 1,2-dichloroethane • Explosion		blockage, check filters periodically • Emergency shut off of process and opening relief system must be install and periodically tested • Installation of evacuation alarm • Remove any source of ignition near by
Reaction	No/ less of	No reaction or minimum reaction occurring	• No or minimum inlet flow • Non ideal temperature • Non ideal pressure • High presence of impurities	• Absence or low amount of product • Zero or low product yield • Zero or low conversion • High impurities of product	• Flow indicators • Temperature indicator • Pressure indicator	• Install a flow indicator control
	More of	Occurrence of more reaction	• Higher reactant flow • Longer time of reactants inside the reactor due to failure in outflow pump or pipe blockage	• Reactor overflow • Increase of temperature inside the reactor • Increase of pressure inside the reactor • Runaway reaction • Formation of side products • Undesired product purity • Failure of outlet pump • Explosion	• Flow indicators • Temperature indicator • Pressure indicator • Inspect lines and pump periodically • Emergency shut off of process • Safety pressure valve release • Evacuation alarm	• Install flow indicator with high alarm (FAH) • Install temperature indicator • Install pressure indicators • Install safety pressure release valve • Install safety evacuation alarm • Remove any source of ignition near by

	Other than	Formation of side reaction	• Non ideal temperature inside the reactor • Non ideal pressure inside the reactor • Non ideal proportion of reactants	• Decrease of product purity • Waste of raw material • Decrease product yield • Decrease reaction conversion • Increase side products production	• Temperature indicator in the reactor • Pressure indicator in the reactor • Flow meters • Inspect lines and pump periodically	• Install a temperature indicator in the reactor • Install a pressure indicator in the reactor • Install a flow indicator for the inlet flow • Periodically inspection of lines, pumps and equipment for deterioration and corrosion
Ph	Higher	Higher Ph	• Side reaction products • Impurities	• Decrease of product yield • Decrease of reaction conversion • Decrease of product purity • Increase rate of corrosion • Damages to equipment	• Ph meters	• Install Ph meters
	Lower	Lower Ph	• Side reaction products • Impurities	• Decrease of product yield • Decrease of reaction conversion • Decrease of product purity • Increase rate of corrosion • Damages to equipment	• Ph meters	• Install Ph meters
Temperature	Less	Low temperature	• Failure of heat exchanger	• Desired reaction may not occur properly	• Temperature indicator	• Install a temperature low alarm (TAL)

			Failure of cooling system	- Decrease of product yield - Decrease of reaction conversion - Decrease of product purity - Decrease of pressure inside reactor	- Control valves for cooling system - Inspect lines and pump periodically	
	More	High temperature	- Failure of heat exchanger system - Failure of cooling system	- Internal temperature of reactor may exceed maximum allowed - Rise of pressure - Failure of equipment - Explosion - Formation of side product - Product yield different than expected - Reaction conversion different than expected - Product purity different than expected	- Temperature indicator - Safety pressure relieve valve - Inspect lines and pump periodically - Control valves for cooling system	- Install a temperature high alarm (TAH) - Install a pressure indicator - Install safety evacuation alarm - Remove any source of ignition near by - Periodically inspection of lines, pumps and equipment for deterioration and corrosion
Pressure	More	High pressure	- Failure of control valve - Failure of temperature control - Failure of pressure control - Pipe blockage - Failure of pump - Runaway reaction - Ph different than desired	- Internal temperature of reactor may exceed maximum safety allowed - Pressure inside the reactor may exceed maximum safety allowed - Failure of equipment - Explosion - Side reaction - Formation of side product - Product yield different than expected	- Temperature indicator - Pressure indicator - Safety pressure relieve valve - Inspect lines and pump periodically	- Install a temperature high alarm (TAH) - Install a pressure indicator - Install a high-pressure alarm (PAH) - Periodically inspection of lines, pumps and equipment for deterioration and corrosion

				• Reaction conversion different than expected • Product purity different than expected		• Install safety evacuation alarm • Remove any source of ignition near by
	Less	Low pressure	• Failure of control valve • Failure of temperature control • Failure of pressure control • Corrosion and leakage in pipeline • Failure of pump • Ph different than desired	• Internal temperature of reactor may not reach desired one • Pressure inside the reactor may not reach desired one • Side reaction • Formation of side product • Product yield different than expected • Reaction conversion different than expected • Product purity different than expected	• Temperature indicator • Pressure indicator • Inspect lines and pump periodically	• Install a temperature low alarm (TAL) • Install a pressure indicator • Install a low-pressure alarm (PAL) • Periodically inspection of lines, pumps and equipment for deterioration and corrosion

HAZOP Study		HCl Column (T-100)				
Intention:		Separate the hydrogen chloride from 1,2-dichloroethane and vinyl chloride. The reboiler is heated by using medium pressure steam and the condenser is cooled by using a refrigerant. This column creates two outlet streams; one of pure hydrogen chloride and one of a mixture of 1,2-dichloroethane and vinyl chloride.				
Parameter	Guide Word	Deviation	Possible Causes	Consequences	Safeguards	Recommendations
Flow	No/ Less off	Inlet flow is reduced or stopped flowing	- Inlet flow leaked out due to pipes corroding, weld/fitting/joints loose or corroded - Inlet fluid pump stopped working due to mechanical problems/ corrosion/ age or cavitation - Pipes became blocked due to biomass build up, mineral build up or corrosion - Failure of control valve	- No properly separation occurs - Decrease of product yield - Release of hazardous material - Increase of pressure inside pipes in case of a blockage - Purity of product may not be same as expected	- Flow meters - Inspect lines and pump periodically - Level indicator	- Install a level indicator with low alarm (FAL) - Periodically inspection of lines, pumps and equipment for deterioration and corrosion
	More	Inlet flow is increased	- Failure of control valve - Pressurized pipelines - Failure in pump control system	- Column is overflowed - Separation doesn't occur as desired - Increased of pressure inside separator - Undesired product purity - Waste of material - Explosion	- Flow meters - Inspect lines and pump periodically - Level indicator	- Install a level indicator with high alarm (FAH) - Periodically inspection of lines, pumps and equipment - Install an emergency shut off system
	Reverse	Reverse flow in pipelines	- Failure of control valves - Failure of pressure controller - Failure of pump	- Outlet flow of the separator is in reverse direction - Decrease in product yield	- Flow meters - Inspect lines and pump periodically	- Install a flow indicator - Periodically inspection of lines, pumps and equipment

				Decrease in reaction conversion Decrease in product purity		
Temperature	Less	Low temperature	- Failure of heat exchanger system - Failure of cooling system	- Desired separation may not occur properly - Decrease of product purity - Decrease of product yield	- Temperature indicator - Control valves for cooling system - Controller for the pump - Inspect lines and pump periodically	Install a temperature low alarm (TAL)
	More	High temperature	- Failure of heat exchanger system - Failure of cooling system	- Internal temperature of separator may exceed maximum allowed - Rise of pressure - Failure of equipment - Explosion - Product yield different than expected - Product purity different than expected	- Temperature indicator - Inspect lines and pump periodically - Control valves for cooling system - Controller for the pump	- Install a temperature high alarm (TAH) - Install a pressure indicator - Install emergency shut off control - Install safety evacuation alarm - Remove any source of ignition near by - Periodically inspection of lines, pumps and equipment for deterioration and corrosion
Pressure	More	High pressure	- Failure of control valve - Failure of temperature control	- Internal temperature of column may exceed maximum safety allowed - Pressure inside the column may exceed	- Temperature indicator - Pressure indicator - Safety pressure relieve valve	- Install a temperature high alarm (TAH) - Install a pressure indicator

			• Failure of pressure control • Pipe blockage • Failure of pump	maximum safety allowed • Failure of equipment • Explosion • Product yield different than expected • Product purity different than expected	• Inspect lines and pump periodically	• Install a high-pressure alarm (PAH) • Periodically inspection of lines, pumps and equipment for deterioration and corrosion • Install safety evacuation alarm • Remove any source of ignition near by
	Less	Low pressure	• Failure of control valve • Failure of temperature control • Failure of pressure control • Corrosion and leakage in pipeline • Failure of pump	• Internal temperature of column may not reach desired one • Pressure inside the column may not reach desired one • Product purity different than expected • Product yield different than expected	• Temperature indicator • Pressure indicator • Inspect lines and pump periodically	• Install a temperature low alarm (TAL) • Install a pressure indicator • Install a low-pressure alarm (PAL) • Periodically inspection of lines, pumps and equipment for deterioration and corrosion

4.0 References

1. <https://www.icis.com/explore/resources/news/2007/11/02/9075707/ethylene-dichloride-edc-production-and-manufacturing-process/>
2. <https://www.osha.gov/laws-regs/regulations/standardnumber/1910/1910.119AppA>
3. <https://www.osha.gov/dsg/annotated-pels/>
4. <https://chemicalsafety.com/sds-search/>
5. <https://www.sepa.org.uk/library/content-search/?q=chlorine&LibGo=Search&page=1>
6. https://www.engineeringtoolbox.com/gases-solubility-water-d_1148.html
7. Material Safety Data Sheet (MSDS) vinyl-chloride
8. “Quantitative Plant Hazard and Risk Assessment”
9. “Hazard & Operability Analysis (HAZOP)”
10. “Process safety 2: Analysis of Process Safety” – CHE 396 Lecture Notes

